Programme	CSBS / EXCP / VLSI	**Year**	S.Y. B.Tech
Course Name and Code	Python Programming Laboratory (316U50L301)	**Semester**	III
Faculty In-charge(s)	Umang Patel, Pragya Gupta, Payal V.	**Academic Year**	2026–27

Distribution of Laboratory CA Marks

Component	Description	Schedule / Coverage	Marks
Component 1	Laboratory Experiments (EXP 1–12)	July–Oct 2026	15
Component 2	Attendance	Entire Semester	5
Component 3	Timely Submission (EXP 1–12)	July–Oct 2026	5
Component 4	On Screen Test (Module 1,2,3.1)	Mid September	15
Component 5	Capstone Project (Entire Syllabus)	End of October	10

Evaluation Rubrics

Laboratory Experiments (15 Marks)

Criterion	Wt.	Level 4	Level 3	Level 2	Level 1
Pre-Lab Preparation & Conceptual Readiness	15%	Arrives with algorithm/ pseudocode fully worked out and explains approach.	Mostly complete algorithm with minor gaps.	Partial plan; needs guidance.	No preparation.
Program Logic & Implementation Correctness	35%	Correct on first attempt; efficient logic.	Correct after minor debugging.	Partially correct with notable errors.	Incorrect or does not execute.
Debugging & Exception Handling	15%	Handles edge cases and exceptions independently.	Handles common errors with minor help.	Basic error handling only.	Cannot debug or handle exceptions.
Output Correctness & Result Interpretation	20%	Correct output with clear interpretation.	Correct output with limited interpretation.	Partially correct output.	Incorrect/no output.
Writeups	15%	Complete writeup with handwritten post-lab and faculty signature.	Complete without handwritten post-lab/ signature.	Partial submission.	Major mistakes/ incomplete.

Attendance Rubric (5 Marks)

Note: Any type of concession will not be considered.

Attendance	Marks
≥75%	5
75–64%	4
65–61%	3
60–55%	2
54–50%	1
<50%	0

Timely Submission Rubric (5 Marks)

Submission Pattern	Marks
All 12 experiments submitted on/before due date	5
10–11 on time	4
8–9 on time	3
6–7 on time	2
3–5 on time	1
0–2 on time	0

On-Screen Test

Marking scheme and solution will be provided after the examination.

Capstone Project Rubrics (10 Marks)

Criterion	Weight	Bloom's Level	CO Mapped
1. Problem Definition & Dataset Selection	10%	Understand / Apply	CO1
2. Code Structure, Modularity & Use of Functions/Modules	15%	Apply	CO1, CO2
3. Data Cleaning & Manipulation (NumPy/Pandas)	20%	Apply / Analyze	CO3
4. Data Visualization (Matplotlib/Seaborn)	20%	Analyze / Create	CO4
5. Statistical Analysis / Forecasting (SciPy/Statsmodels)	20%	Evaluate / Create	CO5
6. Report Quality & Interpretation of Results	10%	Evaluate	CO4, CO5
7. Viva / Project Defense	5%	Understand–Evaluate (all COs)	CO1–CO5

1. Problem Definition & Dataset Selection (CO1)

Level	Descriptor
4 – Excellent	Clear, well-justified problem statement; dataset is appropriate in size/complexity and clearly described.
3 – Good	Reasonable problem statement and dataset; justification present but limited.
2 – Satisfactory	Vague problem statement; dataset appropriateness not discussed.
1 – Poor	No clear problem statement; dataset unsuitable or undocumented.

2. Code Structure, Modularity & Use of Functions/Modules (CO1, CO2)

Level	Descriptor
4 – Excellent	Code is organized into well-named functions/user-defined modules; no repeated logic; easy to follow end-to-end.
3 – Good	Mostly modular with minor repetition or a few oversized functions.
2 – Satisfactory	Some functions used, but significant code duplication or a monolithic script.
1 – Poor	No functions/modules used; single unstructured script.

3. Data Cleaning & Manipulation — NumPy/Pandas (CO3)

Level	Descriptor
4 – Excellent	Missing values, duplicates and outliers handled appropriately and justified; efficient use of Pandas/NumPy operations (groupby, vectorized ops, iloc/loc).
3 – Good	Data cleaned adequately; some operations could be more efficient (e.g., avoidable loops instead of vectorization).
2 – Satisfactory	Basic cleaning done; some missing/duplicate data left unaddressed.
1 – Poor	Little to no data cleaning; frequent errors from unhandled missing/invalid data.

4. Data Visualization — Matplotlib/Seaborn (CO4)

Level	Descriptor
4 – Excellent	Chart types well-chosen for the data/question; plots are labeled, readable, and reveal genuine insight.
3 – Good	Appropriate chart types used; minor labeling/formatting issues.
2 – Satisfactory	Charts present but poorly chosen or hard to interpret (missing labels/titles).
1 – Poor	Few or no meaningful visualizations produced.

5. Statistical Analysis / Forecasting — SciPy/Statsmodels (CO5)

Level	Descriptor
4 – Excellent	Correct statistical test/model chosen and justified (e.g., stationarity check before ARIMA); results correctly interpreted.
3 – Good	Correct method applied; interpretation of results is only partially explained.
2 – Satisfactory	Method applied with some errors (e.g., skipped stationarity check) or minimal interpretation.
1 – Poor	Incorrect or no statistical/forecasting method applied.

6. Report Quality & Interpretation of Results (CO4, CO5)

Level	Descriptor
4 – Excellent	Report clearly connects visualizations/statistics back to the original problem statement with a well-justified conclusion.
3 – Good	Report covers all sections; conclusions are reasonable but not deeply justified.
2 – Satisfactory	Report is present but descriptive only, with weak linkage back to the problem.
1 – Poor	Report missing key sections or shows no connection between analysis and conclusions.

7. Viva / Project Defense (CO1–CO5)

Level	Descriptor
4 – Excellent	Confidently explains every design/analysis choice and can extend the solution to a modified dataset/question on the spot.
3 – Good	Explains most choices correctly with minor prompting.
2 – Satisfactory	Explains only the parts of the project completed independently; struggles with the rest.
1 – Poor	Unable to explain design or analysis choices in their own project.

Date: _____ **Name & Signature of Faculty In-charge(s):**